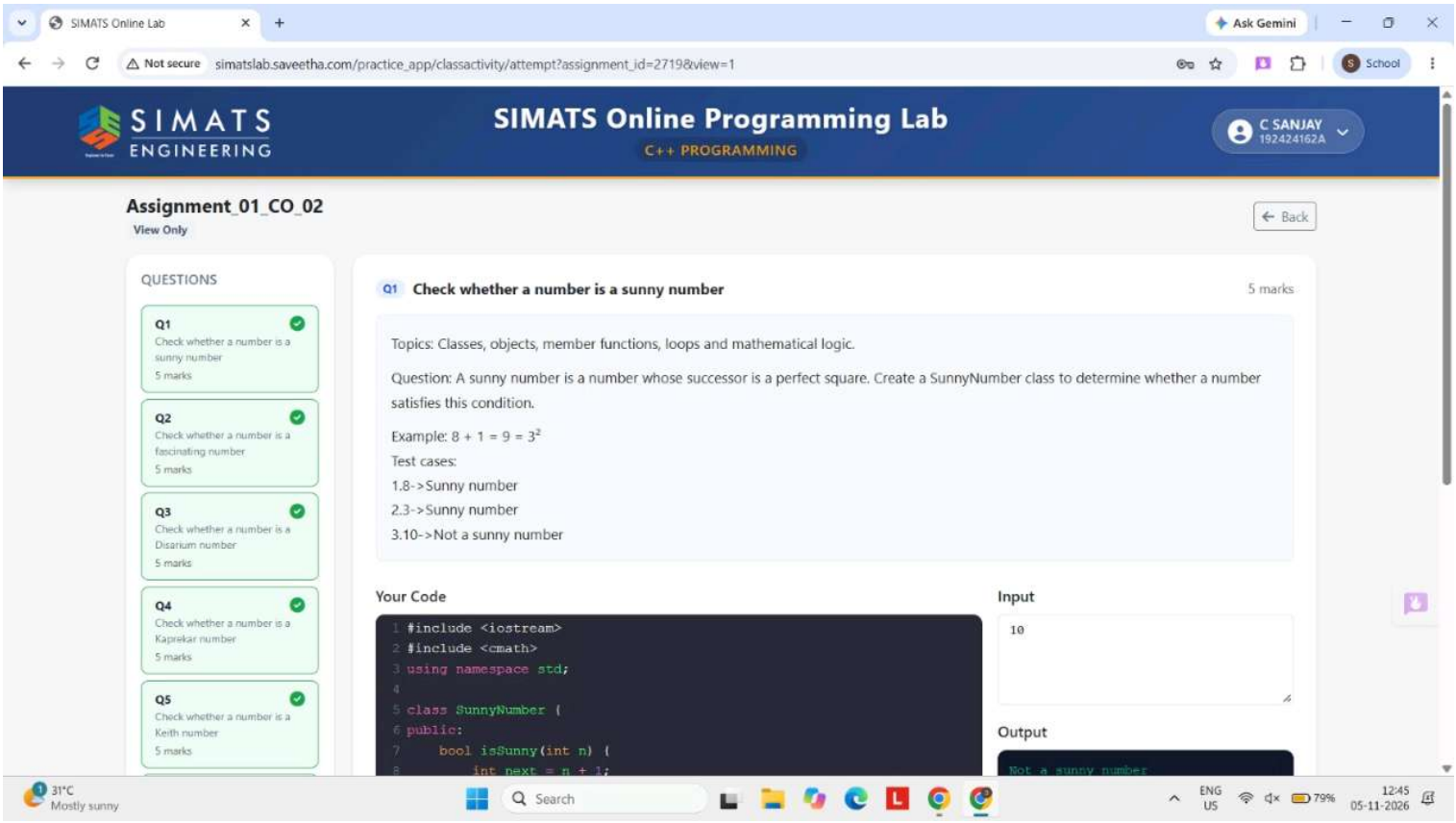

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QUESTIONS
Q1
Check whether a number is a sunny number
5 marks
Q2
Check whether a number is a fascinating number
5 marks
Q3
Check whether a number is a Disarium number
5 marks
Q4
Check whether a number is a Kaprekar number
5 marks
Q5
Check whether a number is a Keith number
5 marks
Q1 Check whether a number is a sunny number
5 marks
Topics: Classes, objects, member functions, loops and mathematical logic.
Question: A sunny number is a number whose successor is a perfect square. Create a SunnyNumber class to determine whether a number satisfies this condition.
Example: 8 + 1 = 9 = 3²
Test cases:
1.8->Sunny number
2.3->Sunny number
3.10->Not a sunny number
Your Code
#include <iostream>
#include <cmath>
using namespace std;
class SunnyNumber {
public:
bool isSunny(int n) {
int next = n + 1;
Input
10
Output
Not a sunny number
31°C
Mostly sunny
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C++ PROGRAMMING

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Check whether a number is a Disarium number
5 marks

Q4
Check whether a number is a Kaprekar number
5 marks

Q5
Check whether a number is a Keith number
5 marks

Q6
Check whether a number is a Trimorphic number
5 marks

6/6 answered

3.10->Not a sunny number

Your Code

```cpp
#include <iostream>
#include <cmath>
using namespace std;

class SunnyNumber {
public:
    bool isSunny(int n) {
        int next = n + 1;
        int root = round(sqrt(next));
        return (root * root == next);
    }
};

int main() {
    int t;
    while (cin >> t) {
        SunnyNumber obj;
        if (obj.isSunny(t))
            cout << "Sunny number" << endl;
        else
            cout << "Not a sunny number" << endl;
    }
    return 0;
}
```

Input

```
10
```

Output

```
Not a sunny number
```

Visible Test Cases

- Test Case 1 ✓
- Test Case 2 ✓
- Test Case 3 ✓

5.00 / 5 marks

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QUESTIONS

- **Q1** Check whether a number is a sunny number — 5 marks
- **Q2** Check whether a number is a fascinating number — 5 marks
- **Q3** Check whether a number is a Disarium number — 5 marks
- **Q4** Check whether a number is a Kaprekar number — 5 marks
- **Q5** Check whether a number is a Keith number — 5 marks
- **Q6** Check whether a number is a Trimorphic number

Q2 Check whether a number is a fascinating number

5 marks

Topics: Classes, member functions, loops, digit extraction, logical validation.

A three-digit number is fascinating if multiplying it by 1, 2, and 3 and concatenating the results produces each digit from 1 to 9 exactly once. Create a NumberAnalyzer class to check the condition.

Example: 192 → 192, 384, 576; together they contain digits 1–9 exactly once.

Test cases:

1.192->Fascinating number

2.219->Fascinating number

3.123->Not a fascinating number

Your Code

```
#include <iostream>
#include <string>
#include <algorithm>
using namespace std;

class NumberAnalyzer {
private:
    int number;

public:
    NumberAnalyzer(int num) {
```

Input

123

Output

Not a fascinating number

31°C Mostly sunny

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Kaprekar number
5 marks

Q5
Check whether a number is a Keith number
5 marks

Q6
Check whether a number is a Trimorphic number
5 marks

6/6 answered

```cpp
#include <string>
#include <algorithm>
using namespace std;

class NumberAnalyzer {
private:
    int number;

public:
    NumberAnalyzer(int num) {
        number = num;
    }

    bool isFascinating() {
        string concatenated = "";
        for (int i = 1; i <= 3; i++) {
            concatenated += to_string(number * i);
        }

        if (concatenated.length() != 9) {
            return false;
        }

        bool seen[10] = {false};
        for (char c : concatenated) {
            int digit = c - '0';
            if (digit == 0 || seen[digit]) {
                return false;
            }
```

Output

```
Not a fascinating number
```

Visible Test Cases

Test Case 1

Test Case 2

Test Case 3

5.00 / 5 marks

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QUESTIONS

- **Q1** Check whether a number is a sunny number — 5 marks
- **Q2** Check whether a number is a fascinating number — 5 marks
- **Q3** Check whether a number is a Disarium number — 5 marks
- **Q4** Check whether a number is a Kaprekar number — 5 marks
- **Q5** Check whether a number is a Keith number — 5 marks
- **Q6** Check whether a number is a Trimorphic number

Q3 Check whether a number is a Disarium number

5 marks

Topics to use: Classes, objects, private data members, public member functions, loops, digit extraction, mathematical calculations.

Question: A number is called a Disarium number if the sum of its digits raised to their respective positions equals the original number. The position count starts from the leftmost digit.

Example: 135

- \(1^1 = 1\)
- \(3^2 = 9\)
- \(5^3 = 125\)
- Sum = \(1 + 9 + 125 = 135\)

Test Cases:

1.175->Disarium number

2.123->Not a Disarium number

3.0->Invalid input

Your Code

```
#include <iostream>
#include <cmath>
#include <string>

using namespace std;
```

Input

```
0
```

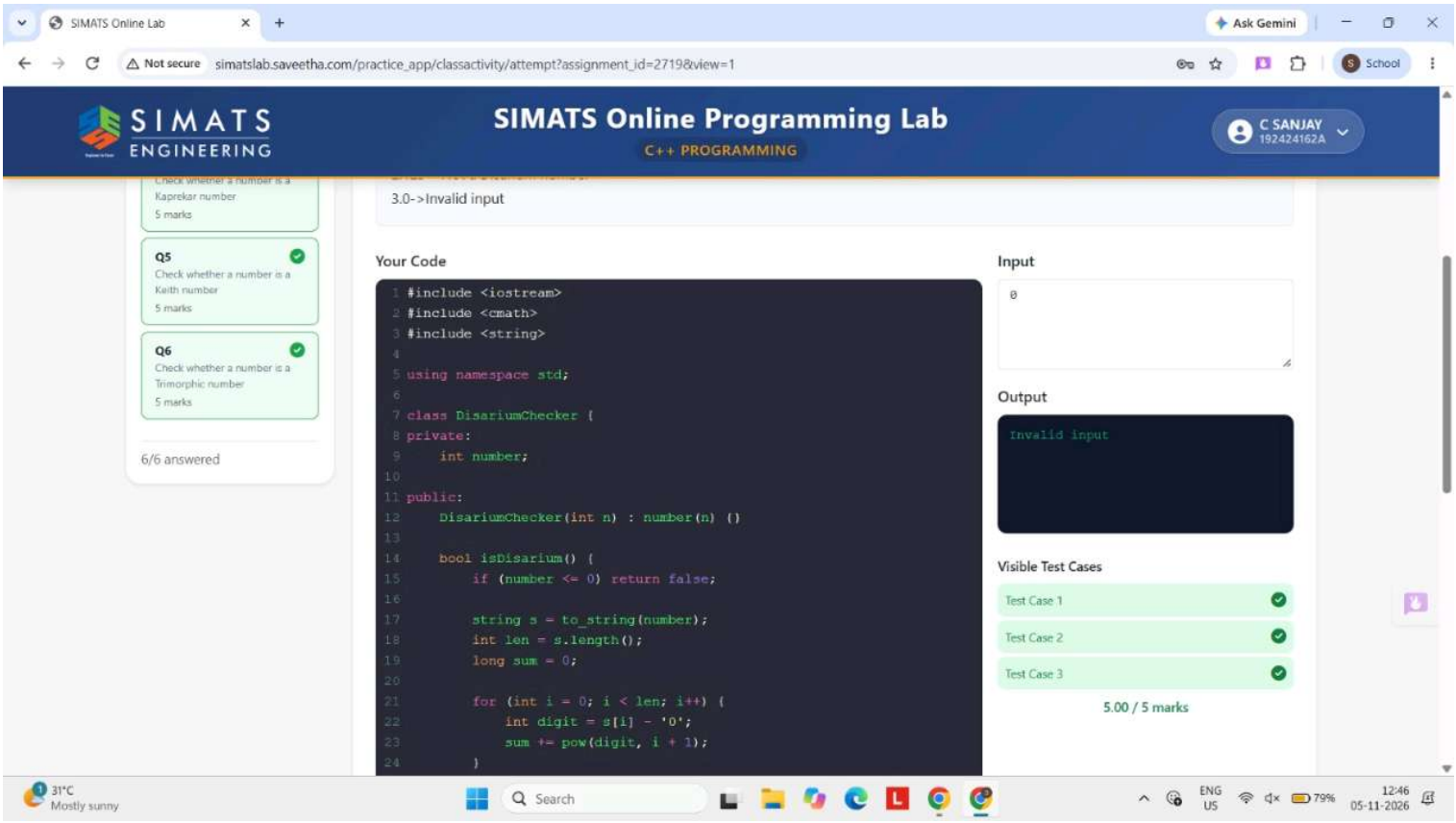
31°C Mostly sunny

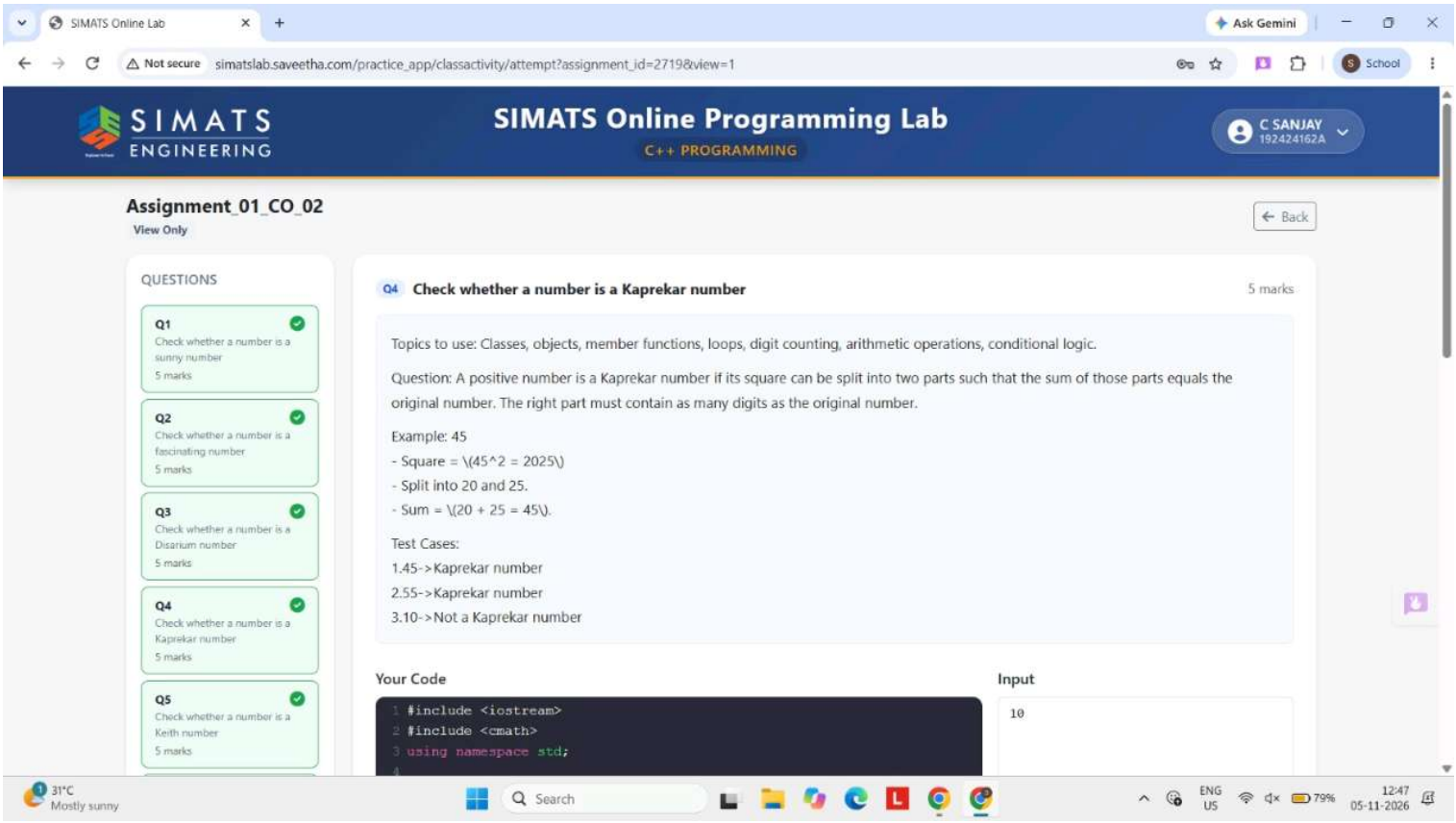
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Kaprekar number
5 marks
Q5
Check whether a number is a Keith number
5 marks
Q6
Check whether a number is a Trimorphic number
5 marks
6/6 answered
3.0->Invalid input
Your Code
#include <iostream>
#include <cmath>
#include <string>
using namespace std;
class DisariumChecker {
private:
int number;
public:
DisariumChecker(int n) : number(n) {}
bool isDisarium() {
if (number <= 0) return false;
string s = to_string(number);
int len = s.length();
long sum = 0;
for (int i = 0; i < len; i++) {
int digit = s[i] - '0';
sum += pow(digit, i + 1);
}
Input
0
Output
Invalid input
Visible Test Cases
Test Case 1
Test Case 2
Test Case 3
5.00 / 5 marks
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Q3
Check whether a number is a Disarium number
5 marks
Q4
Check whether a number is a Kaprekar number
5 marks
Q5
Check whether a number is a Keith number
5 marks
Q4 Check whether a number is a Kaprekar number
5 marks
Topics to use: Classes, objects, member functions, loops, digit counting, arithmetic operations, conditional logic.
Question: A positive number is a Kaprekar number if its square can be split into two parts such that the sum of those parts equals the original number. The right part must contain as many digits as the original number.
Example: 45
- Square = \(45^2 = 2025\)
- Split into 20 and 25.
- Sum = \(20 + 25 = 45\).
Test Cases:
1.45->Kaprekar number
2.55->Kaprekar number
3.10->Not a Kaprekar number
Your Code
#include <iostream>
#include <cmath>
using namespace std;
Input
10
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Q4
Check whether a number is a Kaprekar number
5 marks

Q5
Check whether a number is a Keith number
5 marks

Q6
Check whether a number is a Trimorphic number
5 marks

6/6 answered

3.10->Not a Kaprekar number

Your Code

```cpp
#include <iostream>
#include <cmath>
using namespace std;

class KaprekarChecker
{
public:
    bool isKaprekar(long long n) {
        if (n == 1) return true;

        long long square = n * n;

        int digitCount = 0;
        long long tempN = n;
        while (tempN > 0) {
            digitCount++;
            tempN /= 10;
        }


        long long divisor = 1;
        for (int i = 0; i < digitCount; i++) {
            divisor *= 10;
        }
```

Input

```
10
```

Output

```
Not a Kaprekar number
```

Visible Test Cases

Test Case 1 ✓

Test Case 2 ✓

Test Case 3 ✓

5.00 / 5 marks

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QUESTIONS

Q1
Check whether a number is a sunny number
5 marks

Q2
Check whether a number is a fascinating number
5 marks

Q3
Check whether a number is a Disarium number
5 marks

Q4
Check whether a number is a Kaprekar number
5 marks

Q5
Check whether a number is a Keith number
5 marks

Q5 Check whether a number is a Keith number

5 marks

Topics to use: Classes, objects, member functions, loops, arrays, digit extraction, sequence generation.

Question: A Keith number is a positive integer whose digits form the beginning of a sequence. Each subsequent term is the sum of the previous terms, using as many terms as the original number has digits. If the sequence reaches the original number, it is a Keith number.

Example: 197

Starting digits: 1, 9, 7
- \(1+9+7=17\)
- \(9+7+17=33\)
- \(7+17+33=57\)
- \(17+33+57=107\)
- \(33+57+107=197\)

Therefore, 197 is a Keith number.

Test Cases:
1.742->Keith number
2.14 ->Keith number
3.123->Not a Keith number

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Q5
Check whether a number is a Keith number
5 marks

Q6
Check whether a number is a Trimorphic number
5 marks

6/6 answered

2.14 ->Keith number
3.123->Not a Keith number

Your Code

```
#include <iostream>
#include <vector>
#include <string>
#include <cmath>
using namespace std;

class KeithNumber
{
private:
    int number;
    vector<long long> sequence;

public:
    KeithNumber(int n) : number(n) {}

    bool isKeith() {
        string s = to_string(number);
        int d = s.length();

        if (number < 10) return false;

        for (char c : s) {
            sequence.push_back(c - '0');
```

Input

123

Output

Not a Keith number

Visible Test Cases

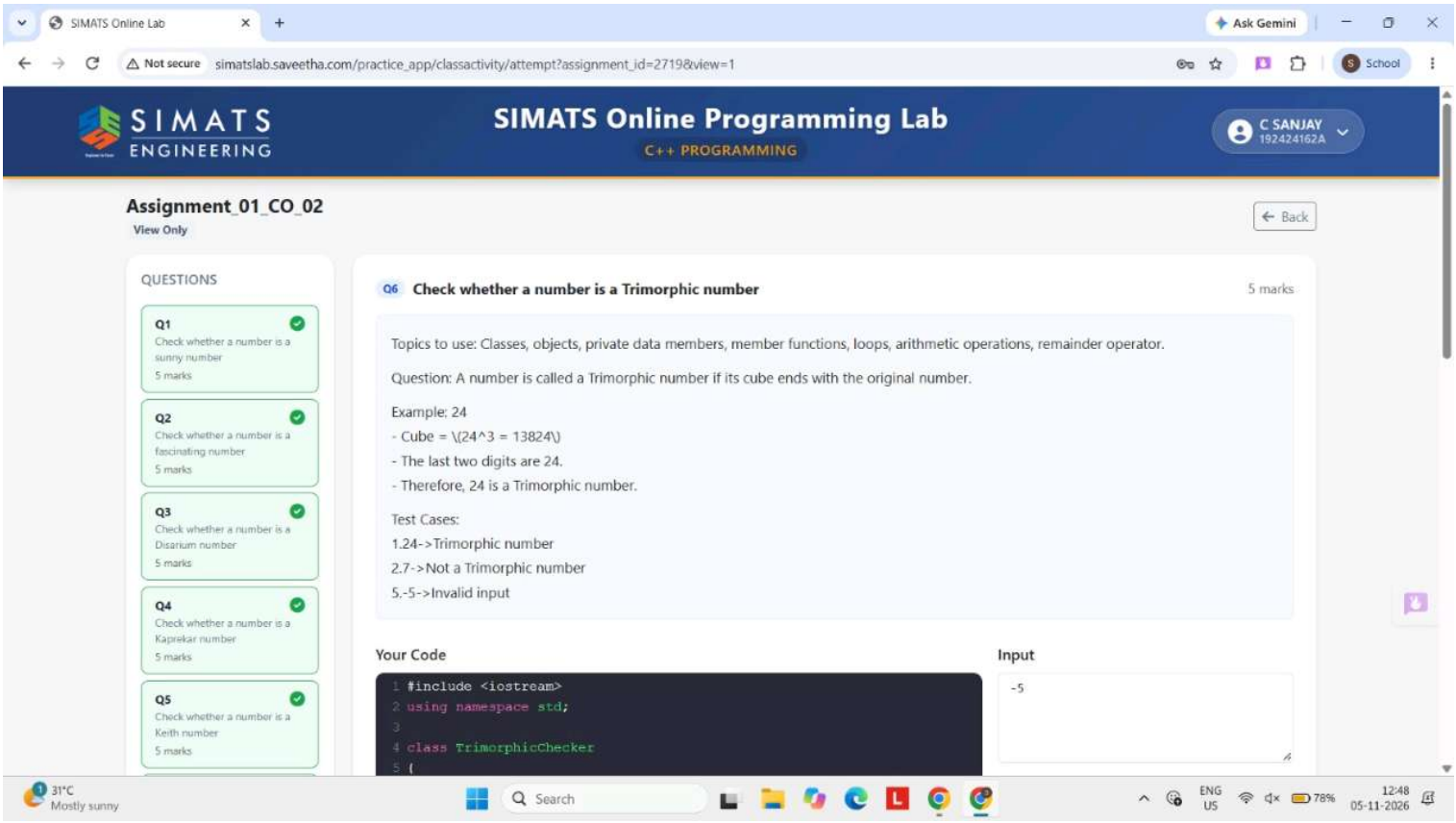
Test Case 1 ✓
Test Case 2 ✓
Test Case 3 ✓

5.00 / 5 marks

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5 marks
Q3
Check whether a number is a Disarium number
5 marks
Q4
Check whether a number is a Kaprekar number
5 marks
Q5
Check whether a number is a Keith number
5 marks
Q6 Check whether a number is a Trimorphic number
5 marks
Topics to use: Classes, objects, private data members, member functions, loops, arithmetic operations, remainder operator.
Question: A number is called a Trimorphic number if its cube ends with the original number.
Example: 24
- Cube = \(24^3 = 13824\)
- The last two digits are 24.
- Therefore, 24 is a Trimorphic number.
Test Cases:
1.24->Trimorphic number
2.7->Not a Trimorphic number
5.-5->Invalid input
Your Code
1 #include <iostream>
2 using namespace std;
3
4 class TrimorphicChecker
5 {
Input
-5
31°C
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Kaprekar number
5 marks

Q5
Check whether a number is a Keith number
5 marks

Q6
Check whether a number is a Trimorphic number
5 marks

6/6 answered

Your Code

```cpp
#include <iostream>
using namespace std;

class TrimorphicChecker
{
private:
    long long number;

public:
    TrimorphicChecker(long long n) : number(n) {}

    void checkTrimorphic() {
        if (number < 0) {
            cout << "Invalid input" << endl;
            return;
        }

        if (number == 0) {
            cout << "Trimorphic number" << endl;
            return;
        }

        long long cube = number * number * number;
        long long temp = number;
        long long divisor = 1;

        while (temp > 0) {
```

Input

```
-5
```

Output

```
Invalid input
```

Visible Test Cases

- Test Case 1 ✓
- Test Case 2 ✓
- Test Case 3 ✓

5.00 / 5 marks

31°C Mostly sunny

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